

TEMPLATE

KEY PROJECT INFORMATION & PROGRAMME DESIGN DOCUMENT (POA-DD)

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VERSION **v. 1.1**

RELATED SUPPORT

- **TEMPLATE GUIDE Key Project Information & PoA Design Document v.1.1**

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Key Project Information

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entity(ies)

KEY PROJECT INFORMATION

GS ID of Programme	GS 11142
Title of Programme:	Installation of High Efficiency Firewood Cookstoves in Sub-Saharan Africa
Start Date of POA	12/05/2021
Date of Design Certification	To be confirmed
POA Period Start Date	16/10/2021
Version number of the PoA-DD	3.0
Completion date of the PoA-DD	21/04/2022
Coordinating/managing entity	C Quest Capital Stoves Asia
Project Participants and any communities involved	Partner organizations in target countries to be mentioned at VPA level.
Host Country (ies)	Republic of Malawi, Republic of Zambia
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Other Requirements applied	Programme of Activity Requirements
Methodology (ies) applied and version number	TPDDTEC (V3.1)
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A

SECTION A. General description of PoA

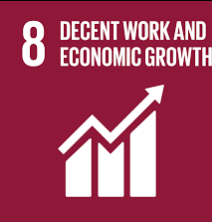
A.1. Purpose and general description of the PoA

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The PoA “Installation of High Efficiency Firewood Cookstoves in Sub-Saharan Africa” will deploy improved cookstoves (ICS) that are substantially more fuel efficient for households. The stoves will replace inefficient three-stone fires thereby reducing the quantity of fuelwood used in cooking meals, thus lowering the GHG emission arising from cooking. Adoption of the ICS will also reduce indoor air pollution and associated health risks. The time spent in collecting firewood will also decrease as the consumption goes down, resulting in more time available for women and girls to engage in other income generating activities or education. Rural communities in Zambia and Malawi are considered as the target population at the start of the PoA. It is also possible to expand the programme to other countries in the region progressively.

Summary of Economic, Social and Environmental impacts of the Project:

Environmental impacts		Reduction of CO2 emissions: The POA will distribute rocket stoves with higher thermal efficiency, therefore using less firewood. This will reduce GHGs emissions from cooking practices.
		Reduced deforestation: Traditional cooking methods use large diameter branches and wood logs cut from trees on farmland and surrounding forests. CQC’s rocket stove will reduce the demand for fuel wood owing to its high thermal efficiency and closed combustion chamber enabling the use of smaller diameter wood such as twigs and thin branches. This will have positive impact on forest degradation and deforestation caused by fuel collection and production in the region.
Social impacts		Improved health conditions: Open fires release toxic particles (PM2.5) that can reach deep into the lungs. Continuously breathing wood smoke can cause severe health impacts such as chronic respiratory disease, acute lower respiratory infections, lung cancer, stroke and cardiovascular disease. Household members using traditional three-stone fires may also suffer burns during cooking. Collecting and carrying heavy bundles of firewood can cause chronic and acute physical ailments.

		Improved burning conditions with CQC's rocket stoves will reduce indoor air pollution and incidences of accidental burns during cooking. This will ensure physical well-being which will be further enhanced by the need to spend less time in collecting firewood.
		Empowerment of women: Health conditions associated with traditional cooking methods will improve particularly for women and young girls with implementation of the project. With reduction in fuelwood requirement, the women and girls will need to make lesser trips for fuel wood collection, thereby reducing the risk of encountering gender-based violence. Furthermore, women will have more time available for leisure, education or opportunities for market employment which can improve their quality of life.
Economic impacts		Creating jobs: Production of stoves parts, distribution/installation, maintenance and marketing activities will create temporary and permanent job opportunities for local people. The employees will acquire new skills and knowledge through trainings provided as well.
		Increased Access to basic services: With ICS, the families will have increased access to basic services one of which is clean cooking technology. Also owing to reduced consumption of fuel wood, the families will have to spend less on purchase of wood fuel thereby resulting in more expendable cash for other activities.

C-Quest Capital Stoves Asia Limited will be the Coordinating Managing Entity (CME) and responsible for coordination of ICS deployment across the geographical boundaries. The CME will be the owner of the carbon credits or VERs generated by the programme. The revenue obtained will be used to improve the affordability and availability of ICS, finance all trainings on construction and use of ICS as well as project monitoring and overall project management. Local organizations and entities will be engaged as implementing partners in the programme to carry out deployment and registration of ICS, training of

the artisans, record keeping and monitoring. The ICS will be distributed on a non-commercial basis by the VPA Implementers.

The PoA is a voluntary action and will be implemented without any mandatory requirements. There are no laws or policies stipulating the adoption of ICS in households in targeted countries.

A.2. Physical/ Geographical boundary of the PoA

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The geographical boundary of PoA will be political boundaries of Republic of Malawi and Republic of Zambia. Coordinates delineating the respective country boundaries are listed below.

Country	Latitude	Longitude
Malawi	13.2543° S	34.3015° E
Zambia	13.1339° S	27.8493° E

A.3. Technologies/measures and eligibility under Gold Standard

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The activities under the proposed PoA will promote improved cook stoves (ICS) that result in reduced fuel consumption and emissions due to cooking and heating water in homes.

For the purpose of this POA, ICS can be classified using the following key characteristics:

- Construction material - Improved Cooking Stoves are commonly made of one or more of the following materials: metal, clay/mud, fired-clay/mud or ceramics and bricks. Classification based on the material helps in selecting an appropriate design on the basis of locally available raw materials, skills for fabrication and necessary production facilities (e.g. centralized/decentralized) in the target area.
- Portability - On this basis, an Improved Cooking Stove can be classified as fixed (in-situ) or portable. Metal and ceramic ICS are normally portable in nature while clay/brick, clay/stone ICS are generally high mass and thus are fixed. Stoves in this category can be further sub-divided into different categories depending on the number of pot holes, e.g., single, double and triple.
- Air circulation- ICS can be Natural Draft Cookstove where the air required for combustion is circulated naturally or Forced Draft Cookstove where an external fan is attached to the stove which runs on external power from mains/battery for proper circulation of air for combustion.

Type	Improved Mud/brick stoves	Improved Metal & Ceramic Stoves
Example	TLC Rocket stove, Plancha Stove etc. 	SSM-S32, WS-2, Kuniokoa etc. 
Portability	Fixed (in-situ)	Portable
Design	Single pot/multi pot	Usually, Single pot
Material	Clay, straw, dung, cement, stone, bricks, and some metal parts with or without chimney	Cast iron or clay with metal exterior body, metal/ceramic pot rests and metal grate
Air Circulation	Usually natural draft but can be converted to forced draft by using an external device such as Jet Flame which consists of a fan that blows powerful jets of air underneath the fire, through the wood, and into the flames creating intense heat and very little smoke.	Usually natural draft but can be converted to forced draft by using an external device such as Jet Flame which consists of a fan that blows powerful jets of air underneath the fire, through the wood, and into the flames creating intense heat and very little smoke.
Stove life	Based on manufacturer's specification/industry standards	Based on manufacturer's specification/industry standards
Efficiency	Usually in the range of 30%-40%.	Usually in the range of 30%-40%.

It is important to note here that models of ICS using different combination of the above categories (construction material, air circulation and portability) may be implemented under this project activity. Specific stove types shall be described in detail in respective VPAs¹.

¹ The models mentioned in the document are for reference. Actual stove model will be defined in respective VPA design document.

It is intended that one of the first stove types to be used under this PoA will be TLC-CQC Rocket Stove. It is an ICS made of locally produced bricks and mortar and high-quality metal parts with an expected lifespan of 10 years. The parts and assembly are shown in the figure below.



Figure. 1. TLC- CQC Rocket stove

The well-cured 5-inch-thick bricks that line the combustion chamber are more durable than metal line stoves. The brick walls are excellent heat sinks and a source of radiative heat to keep combustion chamber temperatures above 600 °C which enables more complete combustion, minimal smoke and a longer simmering duration. The expected thermal efficiency of the stove is 34.5%.

The stainless-steel stick shelf allows air to flow underneath the wood sticks into the combustion chamber resulting in cleaner and more efficient combustion.

The adjustable corrugated steel pot skirt improves the transfer of heat from the fire into the pot, increasing overall energy efficiency of the stove.

Eligibility under GS4GG:

As per 4.1.3 of Principles and Requirements, "A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it." The proposed PoA applies GS approved TPDDTEC methodology (v 3.1) and is among the pre-identified project types that is End use energy efficiency under 3.1.1 (b) of Community Services Activity Requirements (version 1.2).

Eligibility Criteria-Principles and Requirements (V1.2)	PoA's compliance with eligibility Criteria
3.1.1 General eligibility Criteria	
a) Type of project Eligible projects shall include physical action/ implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section.	The PoA falls under Community Activity Project Type. The Methodology applied is GS approved methodology 'Technologies and Practices to Displace Decentralized Thermal Energy Consumption'. All VPAs which will be developed under the present PoA shall belong to 'Community Activity Project' Type and apply GS approved methodology 'Technologies and Practices to Displace Decentralized Thermal Energy Consumption', version 3.1
b) Location of the project Projects may be located in any part of the world.	The PoA presently includes the geographic boundary of Republic of Zambia and Republic of Malawi. Please refer to Section A.2 of Document for details. All VPAs under the present PoA shall implement projects only within the boundary defined by the PoA.
c) Project scale and requirements for avoiding double counting Projects may be developed at any scale (in line with rules, requirements and limitations of Activity Requirements, Impact Quantification Methodologies and Products Requirements). In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects).	The present PoA is a large scale PoA which is in line with Activity, Methodology and Product requirements. Each stove installed under the PoA will be registered by assigning a unique alpha numeric serial number along with GPS coordinates of the end user, his contact details and date of installation. Hence each project stove under the PoA can be identified uniquely thus eliminating any chance of being included in another PoA/project activity under any other carbon standard including Gold Standard. Each VPA will state the location of the Project and provide a range of GPS coordinates and maps to define the Project boundary. This will ensure that no two VPAs under the PoA overlap each other geographically. There could be instances when ICS from another GHG programme being managed by the CME is transferred to VPAs under the present PoA. In such cases, the CME shall clearly mark such stoves and submit proof of zero issuance of credits for the transferred stoves in the concerned monitoring period

	under the other programme to the verifying VVB. Due care must be taken to avoid double counting in such cases.
d) Host Country Requirements Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The present PoA is in compliance with applicable regulations of Malawi and Zambia. Prior to inclusion of a VPA under the present PoA, a self-declaration by CME asserting its compliance with the host country's legal, environmental, ecological and social regulations shall be submitted to VVB.
e) Contact details As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organization (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	The contact details of all project participants can be found in Appendix.1 attached to this document. C-Quest Capital Stoves Asia Limited is a registered company in Malaysia. The contact details of project developer for each VPA will be included in VPA-DDs.
f) Legal Ownership Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	At the time of registration, the end user will be informed that carbon credits generated by the project stoves will be used in improving the affordability of ICS and to promote the programme. In return the end user will be required to surrender their rights to any such credit generated from the project in entirety to the CME. The end user will confirm this via signature/thumb imprint that they are transferring rights to the VERs generated to the CME.
g) Other rights (land, water etc.) well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or	Improved Cookstove project does not require any changes in use of other resources to service the project.

permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	
h) Official Development Assistance (ODA) Declaration All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.	None of the VPAs under the present PoA shall use ODA for implementation of the stoves under it. ODA declaration for each VPA will be submitted prior to its inclusion in the PoA.
i) Section 4.1.1- Climate security & sustainable development Projects shall contribute positively to Climate Security & Sustainable Development.	By installing/distributing ICS, the PoA will result in reduction in fuel wood consumption leading to reduced GHG emission as well as sustainable development of beneficiary communities. The PoA and all associated VPAs comply with all Principles and Requirements. The same will be demonstrated at VPA level.
j) Section 4.1.3- Impact quantification methodology A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it or it's referenced in the Gold Standard Product Requirements. These are published to the Gold Standard website and shall be followed where provided for a given Project type.	All VPAs which will be developed under the present PoA shall apply GS approved methodology 'Technologies and Practices to Displace Decentralized Thermal Energy Consumption', version 3.1.
k) Section 4.1.12- SDG Impact Assessment All Projects shall demonstrate a clear, direct contribution to sustainable development, defined as making demonstrable, positive impacts on at least three Sustainable Development	The PoA's contribution to sustainable development is detailed in Section A.1 of the present document. This will be demonstrated under each VPA-DD as well.

Goals (SDGs), one of which must be SDG 13 (defined herein as Emissions Reductions or Removals and/or Adaptation to climate change). These are referred to as SDG Impacts.	
l) Section 4.1.13- SDG Impact Assessment SDG Impacts shall be a primary effect – an intentional, direct effect of the project and shall not be ‘one off’ or an effect generated in design, construction, distribution, start-up or decommissioning of the Project.	Direct SDG impacts of the programme include: • SDG 13-avoided GHG emission on account of reduced fuelwood consumption • SDG 15-avoided deforestation on account of reduced fuelwood removal from adjoining forests • SDG 3-improved health owing to more efficient combustion of wood fuel resulting in less smoke and soot. • SDG 5-better time management of women and girls resulting from reduced need for wood fuel collection thereby making them available for other finance generating activities • SDG 8- as a result of the PoA, various jobs will be created resulting in economic upliftment of the local communities. • SDG 1- improved access to basic services as well as reduction in fuel requirement leading to reduced expenses on purchase of fuel wood which would finally translate into more dispensable cash for other activities.
m) Section 4.1.14- SDG Impacts Assessment The Project shall identify the potential SDG Impacts provided by the Project by comparing the Project Scenario to the Baseline Scenario. The SDG Impacts shall be demonstrated as making a positive effect beyond what would reasonably be expected to occur in the Baseline Scenario.	In absence of the project activity, none of the 6 SDG impacts mentioned above would be possible.
n) Section 4.1.15- Monitoring Plan The Project shall identify the relevant monitoring indicators and/or monitoring parameters and define the monitoring approach in the Project Design Document to inform future Monitoring Reports.	Monitoring parameters and monitoring approach shall be defined in VPA design document.
o) Section 4.1.19- Safeguarding Principles Assessment Projects shall conduct a Safeguarding Principles Assessment and conform to Gold Standard Safeguarding Principles and Requirements.	Safeguarding Principles Assessment shall be carried out at VPA level (para 7.1.3 of Programme of Activity Requirements, version 1.2)

Eligibility Criteria-Community Services Activity Requirements (V1.2)	PoA's compliance with eligibility Criteria
p) Section 2.1.2- CSA All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	The PoA improves access to improved cookstove technology at highly subsidized rate to end user communities which are households.
q) 3.1.1 (b) End-use energy efficiency: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end. For example, efficient cooking, heating, lighting, etc.	The PoA and all associated VPAs belong to end use energy efficiency category.
r) Section 3.1.2- Project area, boundary and scale: Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements.	Please refer to point 'b' above for details on project boundary. Each project stove shall be identified using a combination of alpha numeric registration code, end user GPS position and contact details as well as installation date making it uniquely identifiable even if other projects of similar nature are implemented within the project boundary. It is a large-scale programme as defined by the Community Services Activity Requirement (version 1.2) and GHG Emissions Reduction & Sequestration Product Requirements as the total energy savings per year for a single VPA is expected to exceed the 180 GWh(th) threshold of small scale.
s) Section 3.1.4- Ownership of distributed units (a) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the	(a) The end user will confirm relinquishing their right to any carbon credit generated from the project via signature/thumb imprint, transferring it in favour of the CME. Please refer to point 'f' above. (b) The transfer of rights of carbon credits has been discussed during the explanation of how carbon finance would be used to support project implementation at the level of local stakeholder consultation.

proofs that end-users are aware of and willing to give up their rights on Products shall be provided. (b) The transfer of Product ownership shall be discussed during local stakeholder consultations for projects	
t) Section 4.1.8- Additionality All projects seeking the issuance of Certified Impact Statements and/or Products shall demonstrate Financial Additionality in accordance with the Principles & Requirements and the applicable Product requirements.	The present PoA involves installation of energy efficient cookstoves and comes under the 'Positive List' as defined under Section 4.1.9 and Annex B of Gold Standard CSA Requirement; version 1.2. According to the list- "Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year" are considered to be auto additional. each of the VPAs, adherence to the energy saving threshold requirement mentioned above shall be demonstrated prior to its inclusion in the PoA.

A.1. Target/Indicator for each of the minimum three SDGs targeted by the POA

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Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	NA	Emission reductions achieved by fuelwood savings at household level
15 Life on Land	15.1. By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in	Amount of firewood saved by the project stoves per year

	particular forests, wetlands, mountains and drylands, in line with obligations under international agreements	
3 Good health and well-being	3.9. By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	Proportion of households who perceive reduced indoor air pollution
5 Gender Equality	5.4. Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Proportion of households who perceive reduced time for collecting wood and cooking Number of women artisans who received training
8 Decent Work and Economic Growth	8.5. By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Number of temporary and permanent jobs created

1 No poverty	1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance Increased access to basic services which includes clean cooking technology
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A.2. Coordinating/managing entity

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C-Quest Capital Stoves Asia is the coordinating/managing entity of the programme.

A.3. Funding sources of PoA

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The project will not receive any public funds.

ODA declaration is provided.

SECTION B. MANAGEMENT SYSTEM AND INCLUSION CRITERIA

B.1. Management System

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The CME will have the overall responsibility of operational and management system of the PoA; which is described as follows:

i. Definitions of roles and responsibilities:

CQC as CME will manage and coordinate activities of the VPA implementers and also provide necessary marketing and promotion assistance to the businesses. The CME will also coordinate the monitoring of the PoA and all the communications with the Gold Standard/ Sustain-Cert.

VPA Implementers will be selected by the CME for the management and coordination of promotion and installation of ICS in host countries. They will also be responsible for monitoring activities once the implementation starts.

VPA Implementers will be selected among NGOs, religious, environmental, social organizations, farmers associations and private, public or governmental entities based on their human capacities and similar experiences in the region. The CME will have an agreement with each VPA implementer establishing roles and responsibilities including:

- Establishing its network for promotion and installation of ICS to the targeted households in rural communities.
- Collecting and recording the ICS as per the management and monitoring plan described.
- Informing the users that emission reductions will be owned by the CME.
- Organizing trainings for installation of stove parts and maintenance.
- Coordinating the monitoring activities of installed ICS.
- After sale services including maintenance and repair.
- Other activities assigned by the CME.

ii. Records of arrangements for training and capacity development for personnel

CME and/or VPA implementer will maintain records of all personnel and arrange trainings for capacity building in order to accomplish the roles defined appointed during the implementation. Key training needs:

- Baseline survey: Perhaps the most important single variable in terms of quantifying VERs is the baseline fuel usage in households and the efficiency of project stoves. The quality of the survey is key in achieving an accurate baseline assessment. For this reason, the CME will provide guidance for interviewers to follow when conducting baseline fuel surveys in homes. This guidance outlines the questions and manner in which the interview should be conducted in order to get the most accurate estimate possible.
- Monitoring: Training, including that of field personnel, is needed to ensure monitoring activities are conducted effectively.
- ICS installation: VPA implementers shall provide evidence of training provided to technicians/instructors/field staff on ICS assembly and installation/distribution.

iii. A procedure for technical review of inclusion of VPAs

CME will review each VPA document and methodically go through each and every eligibility/applicability criterion of the PoA to ensure the VPA meets each requirement with certainty. In cases where there is doubt, the CME will not upload the VPA document until the requirements are met to the CME's satisfaction.

CME will ensure that all proposed VPA implementer's monitoring procedures conform with the monitoring requirements of the PoA, including stove efficiency testing and procedures such as visual inspection and KPT test to check that ICS are still in operation and at what efficiency.

During implementation of the VPA, and as necessary, the CME will ensure all procedures outlined in the PoA are being followed, particularly on stove registration and database updating.

iv. A procedure to avoid double counting

Each ICS included in the programme will be registered with the following information:

- A unique combination of username, community/geographical location, distribution/installation date linked with a unique serial number
- Contact details of the user
- Each VPA will be cross-checked with other VPAs included in the programme as well as project activities registered in other carbon certification schemes to ensure no double-counting takes place.

- The users who are already using an ICS will not be included in the programme.
- There could be instances when ICS from another GHG programme being managed by the CME is transferred to VPAs under the present PoA. In such cases, the CME shall clearly identify such stoves and submit proof of zero issuance of credits for the transferred stoves in the concerned monitoring period under the other programme to the verifying VVB. Failure to do so will result in the particular set of stoves not being included in the project database for the concerned monitoring period.

v. Records and documentation control process for each VPA under the PoA

Each VPA will have a unique project database which will be maintained by the CME. All data will be securely backed up by the CME.

All of records will be collected electronically and stored directly in the monitoring database.

vi. Measures for continuous improvements of the PoA management system

The CME will undertake an annual review of the overall PoA management system, including identifying any problems with stove distribution/installation, stove use once in the homes, monitoring continued stove use and overall database maintenance. This review will ensure that best practices are maintained through the lifetime of the PoA

B.2. Application of methodologies

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GS Methodology: Technologies and Practices to Displace Decentralized Thermal Energy Consumption" (TPDDTEC) Version 3.1 (August 2017) is applied for the PoA.

Applicable GS4GG requirements and guidelines are as follows:

- Programme of Activity Requirements (23/10/2019)
- Community Services Activity Requirements (October 2019)
- Usage Rate Requirements- TPDDTEC (27/10/2020)
- Cookstove Usage Rate Guidelines (27/10/2020)

TPDDTEC (V3.1) methodology defines the following applicability criteria:

Applicability	Justification
The project boundary needs to be clearly identified, and the technologies counted	The registration of each stove includes assigning a unique serial number and

in the project are not included in any other voluntary market or CDM project activity (i.e. no double counting takes place). In some cases, there may be another similar activity within the same target area. Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting.	collecting username, geographic location, contact details and date of installation. The household also receives a registration card with the corresponding serial number. All these details result in a unique identity creation for each stove installed under a particular VPA, enabling stakeholders to identify instances of double counting.
The technologies each have continuous useful energy outputs of less than 150kW per unit (defined as the total useful energy delivered from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	The capacity of each stove will be below the 150kW limit. Calculations demonstrating adherence to same shall be submitted for each model of stove implemented under a VPA at the time of its inclusion. The thermal output of TLC Rocket stoves is included as an example.
Using the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology (e.g. discounted price for the improved technology) and the definitive discontinuity of its use. The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for a good	The CME/VPA implementer shall take steps to encourage complete substitution of baseline stoves with project stoves, however 100 per cent replacement may not be possible. To address this issue, the continued use of baseline stoves will be monitored during annual/biennial surveys and emissions related with the use will be accounted for in the overall emission reduction amount.

understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology.	
The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity. For technology producers and the retailers of the improved technology or the renewable fuel in use, this must be communicated by contract or clear written assertions in the transaction paperwork. If the claimants are not the project technology end users, the end users will need to be informed and notified that they cannot claim for emission reductions from the project	Each user will be informed about the transfer of carbon rights and a carbon credit waiver agreement will be signed between CME and the user at the time of stove registration. The same will be presented to VVB at the time of verification.
Project activities making use of a new biomass feedstock in the project situation (e.g. shift from non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules	N/A

Thermal capacity of the TLC Rocket Stoves is calculated as follows:

Step 1: Estimate the energy consumption in kWh

$$B_{new} = B_{old,i,j} \times \left(\frac{n_{old}}{n_{new}} \right) = 5.5 \frac{\text{ton}}{y} \times \left(\frac{0.1}{0.345} \right) = 1.59 \frac{\text{ton}}{y} \rightarrow 0.00435 \frac{\text{ton}}{\text{day}}$$

$$TE = 0.00435 \frac{\text{ton}}{\text{d}} \times 0.0156 \frac{\text{TJ}}{\text{ton}} \times 277,778 \frac{\text{kWh}}{\text{TJ}} = 18.85 \text{ kWh/d}$$

Step 2: Estimate energy output:

- Based on a survey done by World Bank for Sub-Saharan countries, households spend 3 hours on average for cooking². Accounting that 77%³ of cooking is done with the project stove, number of hours for the project stove is calculated as 2.31 in a day.
- Therefore, the capacity of stove is $18.85 \text{ kWh}/2.31\text{h} = 8.16 \text{ kW}$.

Step 3: Estimate the useful thermal energy output:

- Assuming the thermal efficiency of the stove is 34.5%; the useful output will be $8.16 \text{ kW} * 0.345 = 2.82 \text{ kW}$.
- Emission Reductions will be calculated by the formula:

$$ER_y = \sum_{b,p} (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b,fuel} * (f_{NRB,b,y} * EF_{fuel,CO2} + EF_{fuel,nonCO2}) - \sum LE_{p,y})$$

Where;

$\sum_{b,p}$	Sum over all relevant (baseline b/project p) couples
$N_{p,y}$	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y
$U_{p,y}$	Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage survey (fraction)
$P_{p,b,y}$	Specific fuel savings for an individual technology of project p against an individual technology of baseline b in year y, in tons/day, as derived from the statistical analysis of the data collected from the field tests
$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for firewood, 0.015 TJ/ton)
$f_{NRB,b,y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass
$EF_{fuel,CO2}$	CO ₂ emission factor of the fuel that is substituted or reduced (112 tCO ₂ /TJ for Firewood/Firewood waste)
$EF_{fuel,nonCO2}$	Non-CO ₂ emission factor of the fuel that is reduced

²<https://openknowledge.worldbank.org/bitstream/handle/10986/22521/Climateand0impr000a0landscape0report.pdf?sequence=1&isAllowed=y>
(Page: 154)

³Monitoring report dated 13/08/2019 for CPAs of CDM PoA8060 states that 23% of the households uses a second stove at least once per week as per the baseline study conducted.

$\Sigma LE_{p,y}$ Leakage for project scenario 'p' in year y (tCO₂e/yr)

B.2.1. Multiple technologies/measures

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NA

B.3. Eligibility criteria for inclusion of a VPA in the PoA

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1	Geographic boundary	Each VPA shall be located within the boundaries of the PoA.	ICS registry records
2	Double Counting	A unique numbering system will be adopted for each VPA. All ICS installed shall receive a unique number identifying to which VPA it belongs.	Each ICS included in the programme will be assigned to a VPA and will have a unique combination of customer name and geographical location linked with a unique serial number. For instances where stoves are transferred from CME's managed GHG programme to VPAs under this PoA, a zero-issuance certificate from the GHG programme registry for the transferred stoves shall be submitted to verifying VVB at the time of each verification.
3	Exclusiveness of VPA	VPA shall not be previously: 1. Registered as a project activity with other offset schemes 2. Included as a VPA in any other registered PoA, or deregistered as a VPA of a PoA.	Confirmation by CME
4	Specification of Technology/ Measure	VPAs under the PoA shall install/distribute efficient ICS using firewood. The capacity per unit is limited	The type of stoves replaced and ICS implemented will be

		by 150kW as per the applied TPDDTEC methodology.	described in the specific VPA-DD. Baseline and project Kitchen Performance Tests will be carried out.
5	Start date	The start date of any proposed VPA will be the date of installation of first device under it.	The date on which first ICS installed/distributed will be recorded. Sales invoice or end-user agreement will be provided as a proof.
6	Applicability of methodologies	VPAs shall apply TPDDTEC (v 3.1).	The applicability of the methodology will be demonstrated at VPA level.(Section B.2 of VPA-DD)
7	Additionality	The additionality PoA is demonstrated as per GS4GG Community services activity requirements, Version 1.2 and GHG Emissions Reduction & Sequestration Product Requirements, Version 1.2.	Each VPA will be automatically additional when all evidences are provided: •The project is located in Malawi or Zambia. Total thermal energy savings per unit does not exceed the threshold of 1.8 GWh _{th} .
8	Public funding	Affirmation that there is no diversion of Official Development Assistance (ODA).	Declaration of CME
9	Target group& Distribution Mechanism	Households will be the target group for VPAs. ICSs will be installed by local VPA Implementers/CME representatives on a non-commercial basis.	CME declaration or user registration.
10	Sampling requirements	All requirements as mentioned in TPDDEC, version 3.1 or the Standard: Sampling and surveys for CDM project activities and programme of activities are applicable to VPAs.	Specifications of the sampling methods will be defined at VPA-DDs. VPA Implementers shall follow the management system described at the PoA-DD.

11	Compliance of the technology implemented	The capacity of each ICS will comply with the requirements of TPDDTEC methodology.	VPAs will apply TPDDTEC (v3.1) (Section B.2 of the VPA-DD)
12	SDG Outcomes	Each VPA shall conduct SDG outcomes assessment and comply with the SDG targets identified in the PoA-DD.	VPA-DDs will include SDG outcomes assessment. (Section B.6 of the VPA-DD)
13	Stakeholder Consultation	Each VPA will conduct a local stakeholders' consultation.	Local Stakeholder Consultation Meeting will be conducted for each country.
14	VER Ownership	End users receiving ICSs under the specific VPA contractually cede their rights to claim and own emission reductions to the CME of the PoA.	Carbon right waiver agreement between CME and the user.

SECTION C. DEMONSTRATION OF ADDITIONALITY

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As per GS4GG Community services activity requirements, Version 1.2, Para 4.1.9: Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification:

- (a) Positive list (Annex B of this document)
- (b) Projects located in LDC, SIDS, LLDC
- (c) Microscale projects

The proposed PoA is located in Malawi and Zambia; both fall under the category of LDC and LLDC. The project also complies with the requirement 1.1.3 of Annex B: Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh ($1.8 \text{ GWh}_{\text{th}}$) of energy savings per year or ≤ 600 tonnes of emission reductions per year. This will be demonstrated for each VPA separately.

SECTION D. DURATION OF POA

D.1. Date of first submission of PoA to Gold Standard

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12/05/2021 (First submission of Design Consultation Report)

D.2. Duration of the PoA

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20 years

SECTION E. SAFEGUARDING PRINCIPLES ASSESSMENT

E.1. Justification for Safeguarding Principles Assessment at PoA level

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Safeguarding principles assessment will be done at VPA level.

E.2. Assessment of safeguarding principles, if undertaken at PoA level

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NA

SECTION F. OUTCOME OF STAKEHOLDER CONSULTATIONS

F.1. Justification for stakeholder consultation at PoA Level only

>>NA

F.2. Summary of stakeholder mitigation measures at POA Level

>>NA

F.3. Final Continuous Input / Grievance Mechanism at POA Level

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Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	
GS Contact (mandatory)	help@goldstandard.org
Other	NA

APPENDIX 1 - CONTACT INFORMATION OF COORDINATING/MANAGING ENTITY AND RESPONSIBLE PERSON(S)/ ENTITY(IES)

CME and/or responsible person/ entity	☒ CME ☐ Responsible person/ entity for application of the selected methodology(ies) and, where applicable, the selected standardized baseline(s) to the PoA
Organization	C-Quest Capital Stoves Asia Limited
Street/P.O. Box	Brumby Centre, Lot 42, Jalan Muhibbah
Building	
City	Labuan F.T.
State/Region	
Postcode	8700
Country	Malaysia
Telephone	+6 087 593828
E-mail	cqc-operations@cquestcapital.com
Website	https://cquestcapital.com
Contact person	Ken Newcombe
Title	Director
Salutation	Dr.
Last name	Newcombe
Middle name	Ken

Revision History

Version	Date	Remarks
1.1	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Clarification on POA level LSC and Safeguard Principles Assessment Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption